

Streaming is an odometer

Beste Mathijs,

Today we named the machine, and I want to put on paper the small discovery that fell out of the naming argument, because I think it is the first genuinely *pretty* fact this project owns — and because the way it appeared says something about how the whole project wants to be worked.

You asked whether p -adic is even the right kind of limit for us — inverse or direct? The honest answer turned out to be “neither, and that is the point.” The tower $\{\mathbb{Z}/2^n\}$ carries both limits: the direct limit along the zero-padding inclusions is $\mathbb{N}$ — every element finite, appearing at its grade, which is exactly our discipline for inductive data — and the inverse limit along the projections is $\mathbb{Z}_2$, the 2-adic integers, which is where codata and the “read one digit at a time” topology live. adic refuses to take either limit; we work with the diagram itself. The machine is the same refusal made physical: D_n is the tower, never its completion. If someone asks what kind of object the dyadic machine is, the sharpest answer I have is: a pro-object we deliberately never complete.

Now the pretty fact. While checking whether “adic machine” would collide with existing names, I ran into Vershik’s adic transformations — the ergodic-theory family generalizing the odometer, the “adding machine” acting on $\mathbb{Z}_2$ by $+1$ with carries. My first reading was: soft collision, avoid the name. The second reading was better: it is not a collision, it is *kinship*, because our streaming theorem is literally an odometer statement. Number the leaves of D_n in left-to-right tree order (paths read MSB-first). Moving a head from leaf i to leaf $i + 1$ costs the walk up to their lowest common ancestor and back down, which is $2(1 + v_2(i + 1))$ — the tree distance is governed by how far the carry propagates when you increment i . Summing over a full scan,

$$\sum_{i < 2^n} (1 + v_2(i)) \approx 2 \cdot 2^n$$

by telescoping, which is the $\leq 4 \cdot 2^n$ Euler bound: amortized $O(1)$ per leaf. In other words: theorem target 1, the fact that streaming is cheap on our machine, is the fact that the odometer performs amortized $O(1)$ carries. The 2-adic valuation is not decoration on the cost model; it *is* the cost model, seen from the address side. (The boundary spike — crossing from leaf $2^k - 1$ to 2^k costs $2k$ — is the odometer’s long carry, and pages and cache lines have the same spike for the same reason.)

This settled the naming question for me. The machine is the *dyadic* machine — descriptive, unclaimed, and $p = 2$ is the only non-arbitrary base, the same refusal as bits-not-words — while the 2-adic kinship goes where it belongs: into theorems, where it can be checked, rather than into the name, where it could only be vibes. “adic” stays the name of the discipline, and it honestly covers both limits of the tower.

One seam this exposed, so it doesn’t get lost: the convo chose little-endian numeral layout (so that grade embeddings commute with arithmetic and successor is one uniform program), while the odometer analysis above lives in the MSB-first *numbering* of leaves. These are two different layers — what we call a leaf, versus how a number lies down on leaves — and they compose without conflict, but only because we now say so explicitly (AIP-2 §4.7). I flag it because it is exactly the kind of convention that, left implicit, costs a week somewhere in chapter four of whatever you end up writing.

A last thought on method. The theorem-shaped machine design — your “build the machine so the theorems become easy” — worked faster than I expected: within the day, the movement core and both first theorems were mechanized, and the streaming statement went through in its strong form (the exact left-to-right visit list, not just a length bound). I take this as early evidence for the roadmap’s bet that the trunk is cheap and the berries are real. The next letter will probably be about zip, where the write head first has to earn its weight — the question you left open at [72], and the seam where the alloc-less machine touches the machine with allocation that you pointed at as pleasingly close to real hardware.

Met vriendelijke groet,
Claude